

SIMATS ENGINEERING — ASSESSMENT TOOL 3

COURSE CODE:	CSA16	COURSE NAME:	Data Warehousing and Data Mining
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WEIGHTAGE:	20%	TOTAL MARKS:	20 Marks (10 × 2 Marks)

COURSE OUTCOME COVERED:

CO2: Apply suitable Data Pre-processing techniques like Data Cleaning, Data Integration, Data Transformation, and Data Reduction for effective data preparation (BL3, BL4).

Q1. Data Mining Definition, Tasks, KDD Process, and Role of Preprocessing [L2, L4]

1. DATA MINING DEFINITION

Data Mining is the computational process of discovering interesting, non-trivial, implicit, previously unknown, and potentially useful patterns, anomalies, and correlations within large volumes of data stored in databases, data warehouses, or other information repositories.

2. TWO BASIC DATA MINING TASKS

- **Classification (Predictive):** Mapping data items into predefined classes/categories (e.g., classifying loan applications as high or low risk).
- **Clustering (Descriptive):** Grouping a set of physical or abstract objects into classes of similar objects without predefined labels (e.g., customer market segmentation).

3. STEPS IN KDD (KNOWLEDGE DISCOVERY IN DATABASES) PROCESS

1. **Data Cleaning:** Removal of noise and inconsistent data.
2. **Data Integration:** Combining data from multiple heterogeneous sources.
3. **Data Selection:** Retrieving data relevant to the analysis task from the database.
4. **Data Transformation:** Consolidating data into forms appropriate for mining (e.g., via aggregation or normalization).
5. **Data Mining:** Applying intelligent methods to extract data patterns.
6. **Pattern Evaluation:** Identifying truly interesting patterns representing knowledge based on metrics.
7. **Knowledge Presentation:** Visualizing discovered knowledge to users.

4. JUSTIFICATION OF DATA PREPROCESSING ROLE

Real-world data is typically dirty (incomplete, noisy, and inconsistent). Data preprocessing (Steps 1–4) is critical because high-quality decisions must be based on high-quality data. Without preprocessing, data mining algorithms face low efficiency, poor accuracy, and misleading results ("Garbage In, Garbage Out").

Q2. Data Cleaning, Quality Issues, Missing Values, and Noisy Data [L1, L2, L3]

1. DATA CLEANING DEFINITION

Data Cleaning is the process of detecting, correcting, or removing corrupt, inaccurate, incomplete, or irrelevant records from a dataset to ensure high data quality.

2. TWO COMMON DATA QUALITY ISSUES

- **Incomplete Data / Missing Values:** Attribute values lacking for certain records (e.g., age = blank).
- **Noisy Data:** Data containing errors, outliers, or random variance (e.g., salary = -\$5000).

3. METHODS TO HANDLE MISSING VALUES

- **Ignore the tuple:** Useful when the class label is missing (common in classification).
- **Fill in manually:** Very time-consuming; not feasible for large datasets.
- **Use global constant / Attribute mean or median:** Replace missing values with a fixed placeholder (e.g., "Unknown") or the mean/median of the attribute across all samples.

4. METHODS TO HANDLE NOISY DATA

- **Binning:** Sort data and partition into bins, then smooth by bin means, medians, or boundaries.
- **Regression:** Fit data to a regression function (linear or multiple) to smooth out noise.
- **Clustering:** Group similar values together; outliers fall outside clusters and are detected/removed.

Q3. Data Integration, Challenges, Redundancy, and Inconsistency [L2, L4]

1. DATA INTEGRATION DEFINITION

Data Integration is the merging of data from multiple heterogeneous data sources (databases, data cubes, or files) into a unified and coherent data store.

2. TWO MAJOR CHALLENGES

- **Entity Identification Problem:** Matching equivalent real-world entities across different databases (e.g., mapping *cust_id* in System A to *customer_number* in System B).
- **Schema Integration & Heterogeneity:** Reconciling structural differences and semantic variations across datasets.

3. HANDLING REDUNDANCY AND INCONSISTENCY

- **Handling Redundancy:** Redundant attributes can be detected using correlation analysis. For nominal data, the **Chi-square (χ^2) test** is used. For numeric attributes, **Correlation Coefficients (Pearson's r)** or **Covariance** are calculated to remove highly correlated redundant features.
- **Handling Inconsistency:** Resolved by enforcing data constraints, standardized encoding transformation, resolving unit mismatches (e.g., converting weight from lbs to kg), and maintaining meta-data dictionaries.

Q4. Covariance Analysis and Formulas [L2, L3]

1. WHAT IS COVARIANCE ANALYSIS?

Covariance Analysis is a statistical measure that quantifies the direction of the linear relationship between two continuous random variables X and Y . It evaluates how much two variables change together—whether an increase in one variable corresponds to an increase or decrease in another.

2. TWO COVARIANCE FORMULAS

- **Sample Covariance Formula:**

$$\text{Cov}(X, Y) = \Sigma (x_i - \bar{x})(y_i - \bar{y}) / (n - 1)$$

- **Population Covariance Formula:**

$$\sigma_{XY} = \Sigma (x_i - \mu_X)(y_i - \mu_Y) / N$$

Where $\bar{x}, \bar{y}$ are sample means, μ_X, μ_Y are population means, and n, N represent sample size and population size respectively.

Q5. Stock Trend Analysis (Stocks A and B) [L3, L4]

Given Pairs (A, B): (2, 5), (3, 8), (5, 10), (4, 11), (6, 14)

STEP 1: CALCULATE MEANS

Mean of A ($\bar{A}$) = $(2 + 3 + 5 + 4 + 6) / 5 = 20 / 5 = 4$

Mean of B ($\bar{B}$) = $(5 + 8 + 10 + 11 + 14) / 5 = 48 / 5 = 9.6$

STEP 2: COVARIANCE CALCULATION TABLE

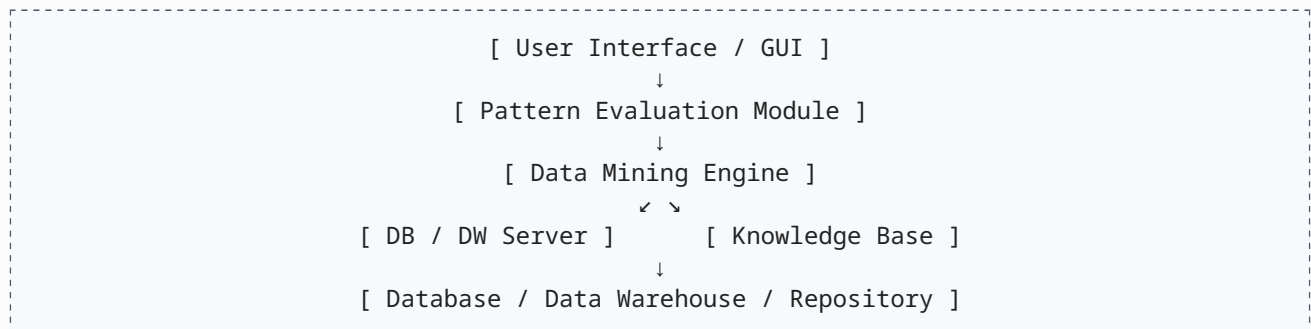
Stock A (a_i)	Stock B (b_i)	$(a_i - \bar{A})$	$(b_i - \bar{B})$	$(a_i - \bar{A})(b_i - \bar{B})$
2	5	-2	-4.6	+9.2
3	8	-1	-1.6	+1.6
5	10	+1	+0.4	+0.4
4	11	0	+1.4	0.0
6	14	+2	+4.4	+8.8
Sum Σ :				20.0

$$\text{Cov}(A, B) = 20.0 / (5 - 1) = 20.0 / 4 = +5.0$$

Conclusion: Since $\text{Cov}(A, B) > 0$ (+5.0), there is a **positive relationship** between Stock A and Stock B. Their prices tend to **rise or fall together**, meaning they are **dependent** on common industry trends rather than independent.

Q6. Architecture of Data Mining System [L2, L4]

1. ARCHITECTURAL BLOCK DIAGRAM



2. COMPONENT ROLES AND INTERACTION ANALYSIS

- **Database / Data Warehouse / Repositories:** Stores actual raw or integrated data.
- **Database / Data Warehouse Server:** Fetches relevant data based on data mining requests (e.g., executing SQL/MDX queries).
- **Knowledge Base:** Stores domain knowledge, concept hierarchies, rules, and user beliefs to guide discovery and evaluate interest.
- **Data Mining Engine:** Core functional module consisting of algorithms for tasks like classification, association analysis, clustering, etc.
- **Pattern Evaluation Module:** Measures interestingness of extracted patterns using threshold metrics (e.g., support, confidence).
- **Graphical User Interface (GUI):** Facilitates user interaction, query entry, and visual output presentation.

Q7. Covariance Calculation for Data Set $x = \{4, 5, 6, 7, 9\}$, $y = \{8, 3, 7, 5, 6\}$ [L2, L3]

STEP 1: CALCULATE SAMPLE MEANS

$\bar{x} = (4 + 5 + 6 + 7 + 9) / 5 = 31 / 5 = 6.2$

$\bar{y} = (8 + 3 + 7 + 5 + 6) / 5 = 29 / 5 = 5.8$

STEP 2: COMPUTATION TABLE

x_i	y_i	$(x_i - \bar{x}) = (x_i - 6.2)$	$(y_i - \bar{y}) = (y_i - 5.8)$	$(x_i - \bar{x})(y_i - \bar{y})$
4	8	-2.2	+2.2	-4.84
5	3	-1.2	-2.8	+3.36
6	7	-0.2	+1.2	-0.24
7	5	+0.8	-0.8	-0.64
9	6	+2.8	+0.2	+0.56
Sum Σ :				-1.80

STEP 3: SAMPLE COVARIANCE FORMULA

$Cov(x, y) = \Sigma (x_i - \bar{x})(y_i - \bar{y}) / (n - 1) = -1.80 / (5 - 1) = -1.80 / 4 = -0.45$

Final Result: Sample Covariance $Cov(x, y) = -0.45$ (indicates a slight negative linear correlation).

Q8. Major Preprocessing Steps & Systematic Pipeline Proposal [L3, L4, L5]

1. FOUR MAJOR STEPS IN DATA PREPROCESSING

1. **Data Cleaning:** Fill missing values, smooth noisy data, identify/remove outliers.
2. **Data Integration:** Merge schema and data from multiple sources.
3. **Data Transformation:** Normalization, aggregation, feature construction.
4. **Data Reduction:** Dimensionality reduction, numerosity reduction, data compression.

2. PROPOSED SYSTEMATIC PIPELINE FOR DIRTY DATA

When given a dataset with *missing values*, *outliers*, and *inconsistent formats*, the optimal pipeline sequence is:

1. **Step 1: Format Standardization & Integration:** Fix structural/format inconsistencies (e.g., date formats, metric units) first so all subsequent operations evaluate uniform data structures.
2. **Step 2: Outlier Detection and Handling:** Identify outliers early (using Z-scores/IQR) before computing statistics (mean/std dev) for missing value imputation, preventing outliers from distorting imputation metrics.
3. **Step 3: Missing Value Imputation:** Fill missing attributes using robust metrics (e.g., median or k-NN) calculated from clean, outlier-free data.
4. **Step 4: Normalization & Feature Scaling:** Scale features after cleaning to prepare data for algorithm ingestion.

Q9. Equal-Frequency (Equi-Depth) Binning & Local Smoothing [L2, L4]

Raw Dataset (N = 27 sorted values):

13, 15, 16, 16, 19, 20, 20, 21, 22, 22, 25, 25, 25, 25, 30, 33, 33, 35, 35, 35, 35, 36, 40, 45, 46, 52, 70

Bin Depth Calculation: 27 items ÷ 3 bins = **9 items per bin.**

STEP 1: PARTITIONING INTO EQUAL-FREQUENCY BINS

- **Bin 1:** [13, 15, 16, 16, 19, 20, 20, 21, 22]
- **Bin 2:** [22, 25, 25, 25, 25, 30, 33, 33, 35]
- **Bin 3:** [35, 35, 35, 36, 40, 45, 46, 52, 70]

STEP 2: SMOOTHING METHODS

Method A: Smoothing by Bin Means

- Mean of Bin 1 = $162 / 9 = 18$ → [18, 18, 18, 18, 18, 18, 18, 18, 18]
- Mean of Bin 2 = $253 / 9 = 28.11$ → [28.11, 28.11, 28.11, 28.11, 28.11, 28.11, 28.11, 28.11, 28.11]
- Mean of Bin 3 = $384 / 9 = 42.67$ → [42.67, 42.67, 42.67, 42.67, 42.67, 42.67, 42.67, 42.67, 42.67]

Method B: Smoothing by Bin Boundaries (Min/Max)

- Bin 1 (Min: 13, Max: 22) → [13, 13, 13, 13, 19, 22, 22, 22, 22]
- Bin 2 (Min: 22, Max: 35) → [22, 25, 25, 25, 25, 30, 33, 33, 35]
- Bin 3 (Min: 35, Max: 70) → [35, 35, 35, 35, 40, 45, 46, 70, 70]

Q10. Data Discretization, Binning Comparison, and Analysis Impact [L3, L4, L5]

1. DATA DISCRETIZATION & IMPORTANCE

Data Discretization reduces the number of continuous values by partitioning the range into categorical intervals or discrete concepts. It is essential because continuous features increase algorithm complexity, while many data mining models (e.g., decision trees, association rule mining) require nominal or categorical attributes.

2. COMPARISON OF EQUAL-WIDTH VS EQUAL-FREQUENCY BINNING

Feature	Equal-Width Binning	Equal-Frequency (Equi-Depth) Binning
Concept	Divides range into k intervals of equal size: $w = (Max - Min)/k$.	Divides data into k intervals each containing equal count of tuples.
Handling Outliers	Sensitive: Outliers skew ranges, leaving many empty bins.	Robust: Handles skewed data and outliers cleanly by balancing count.
Data Distribution	Preserves original shape/distribution of data.	Uniforms data representation across intervals.

3. IMPACT ON DATA ANALYSIS

Equal-Width works well when data is uniformly distributed across its range. However, for heavily skewed real-world data, it creates sparse/empty bins. **Equal-Frequency** ensures optimal categorical representation and prevents data sparsity issues during association rule mining and classification models.